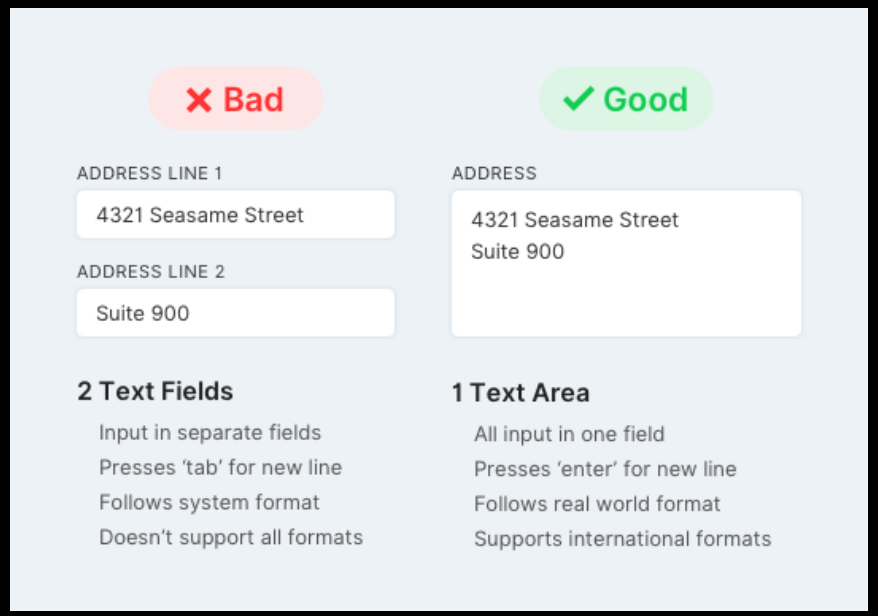
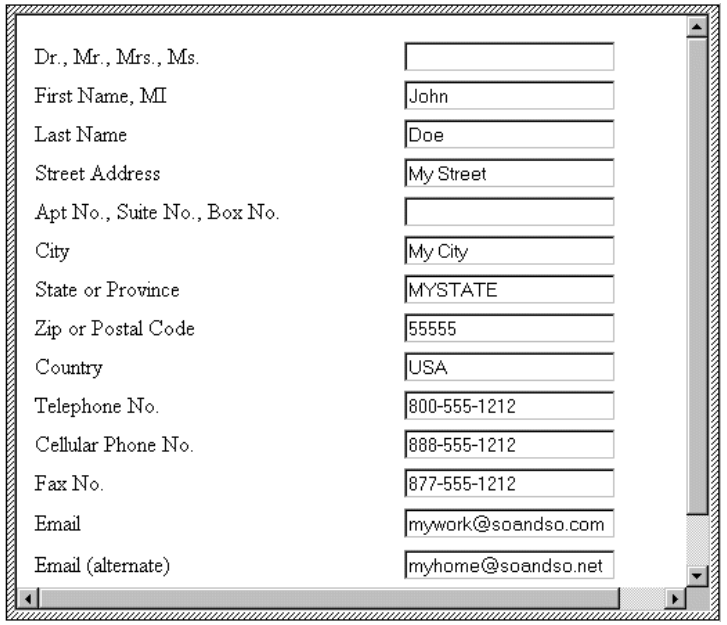
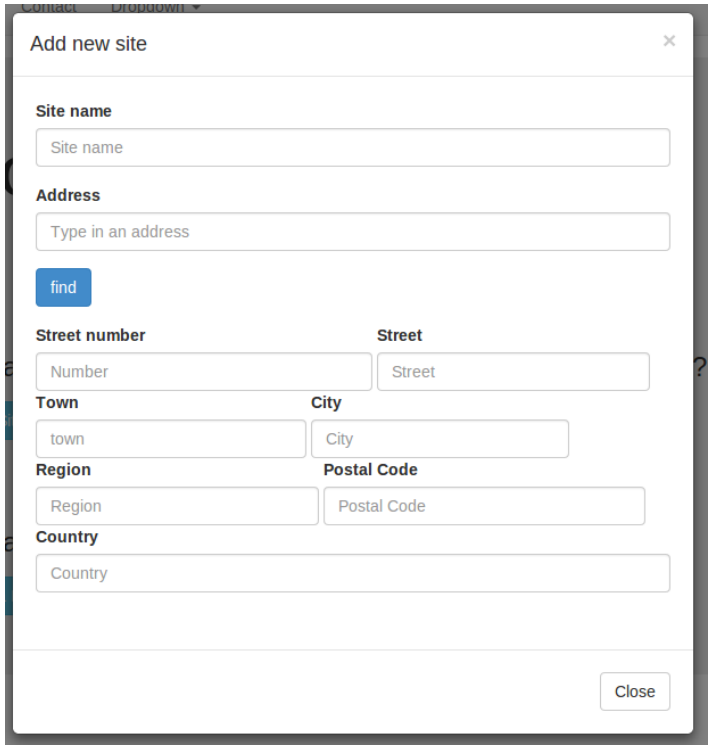
ADDRESS SEGMENTATION

Trần Tuấn Anh

1. PROBLEM DEFINITION: Address representation







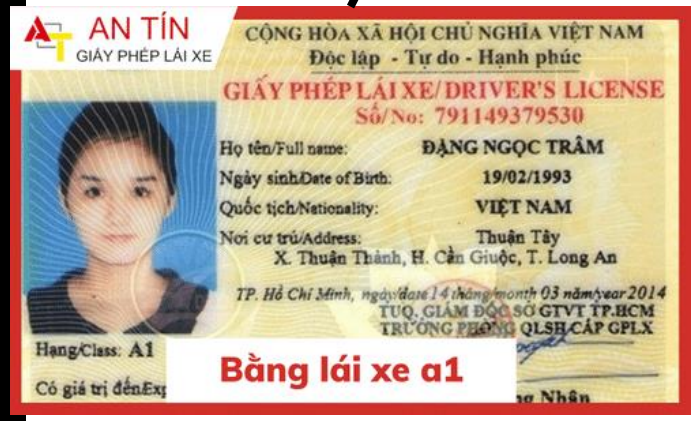
Clarity of information

1. PROBLEM DEFINITION: Human Reading and representation



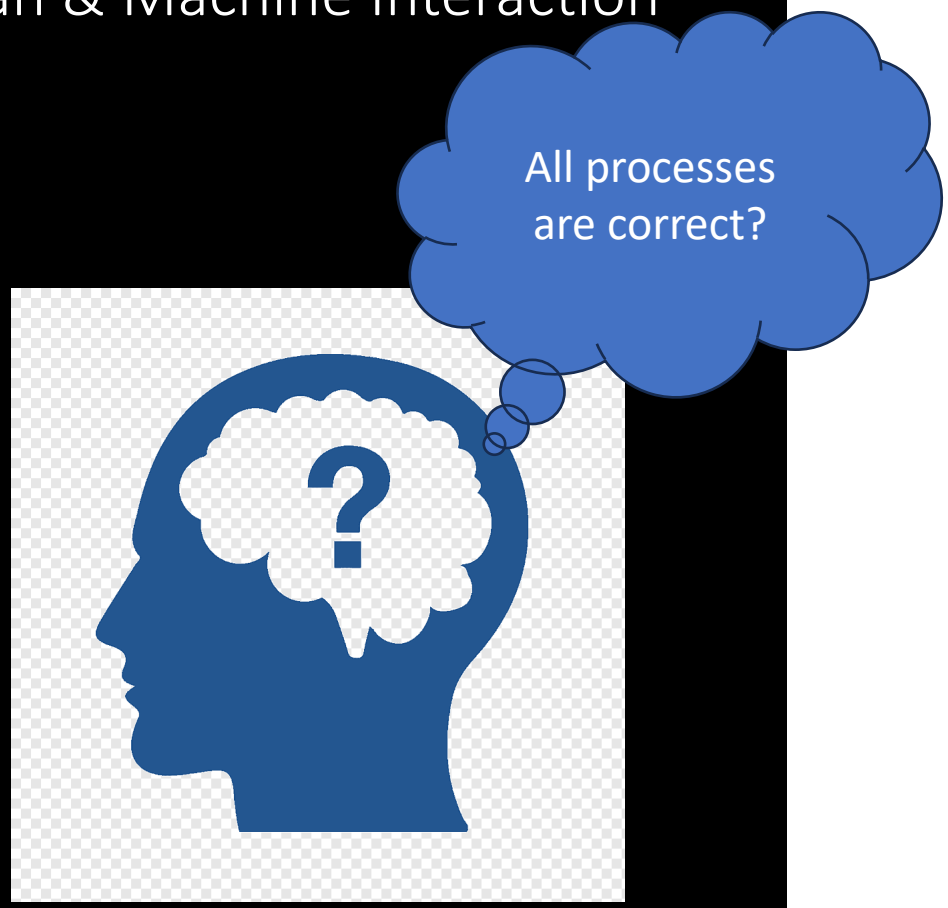


HUMAN READING & UNDERSTANDING: confusing rules





1. PROBLEM DEFINITION: Human & Machine Interaction





Detect address area

OCR

Classify address

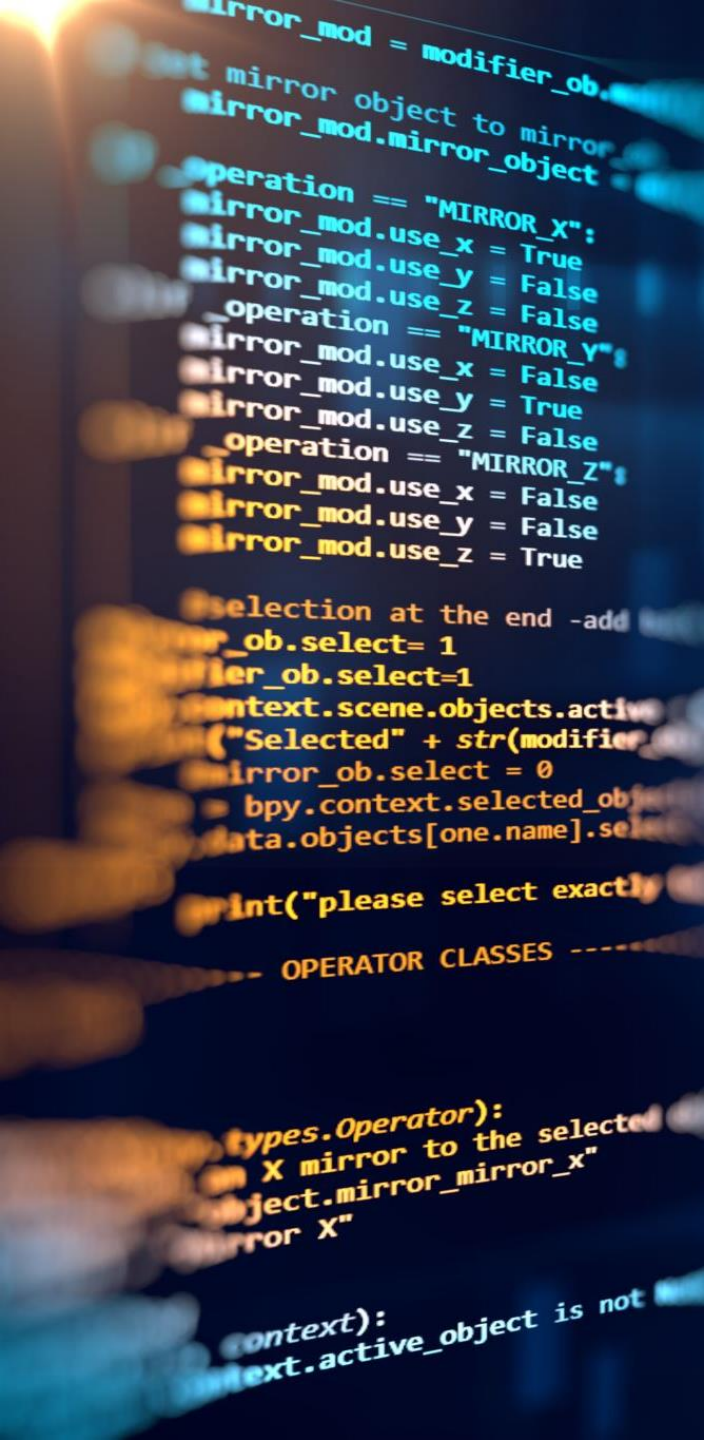
1. PROBLEM DEFINITION:
Input vs Output



OCR:

X. Thuận Thành, H. Cần Giuộc, T. Long An
Thuận Thanh, HCần Giuộc, Tlong An
Thuận Thành, H Cần Giuộc T. Long An
X Thuận Thành H. Cần Giuộc, Long An
X ThuanThanh H. Can Giuoc, Long An
……..

Address classification:
Xã: Thuận Thành
Huyện: Cần Giuộc
Tỉnh: Long An



1. PROBLEM DEFINITION: Input vs Output

Input:

Text string

Database for correction (not formal)

List of Street,

List of Ward,

List of Province

Output:

Street

Ward

Province

Example 1:

Input: 268 Lý Thường Kiệt, Diên Hồng, TP. HCM

Database
Street: Lý Thường Kiệt, Sư Vạn Hạnh, 3/2

Ward: Diên Hồng, Hòa Hưng, Con cò bé bé

Province: TP. HCM, Hà Nội, Vũng Tàu

Output:

Street: Lý Thường Kiệt

Ward: Diên Hồng

Province: TP. HCM

Example 1:

Input: 268 Lý Thường Kiệt, Diên Hồng, TP. HCM

Database
Street: LTK, Sư Vạn Hạnh, 3/2

Ward: Hòa Hưng, Con cò bé bé

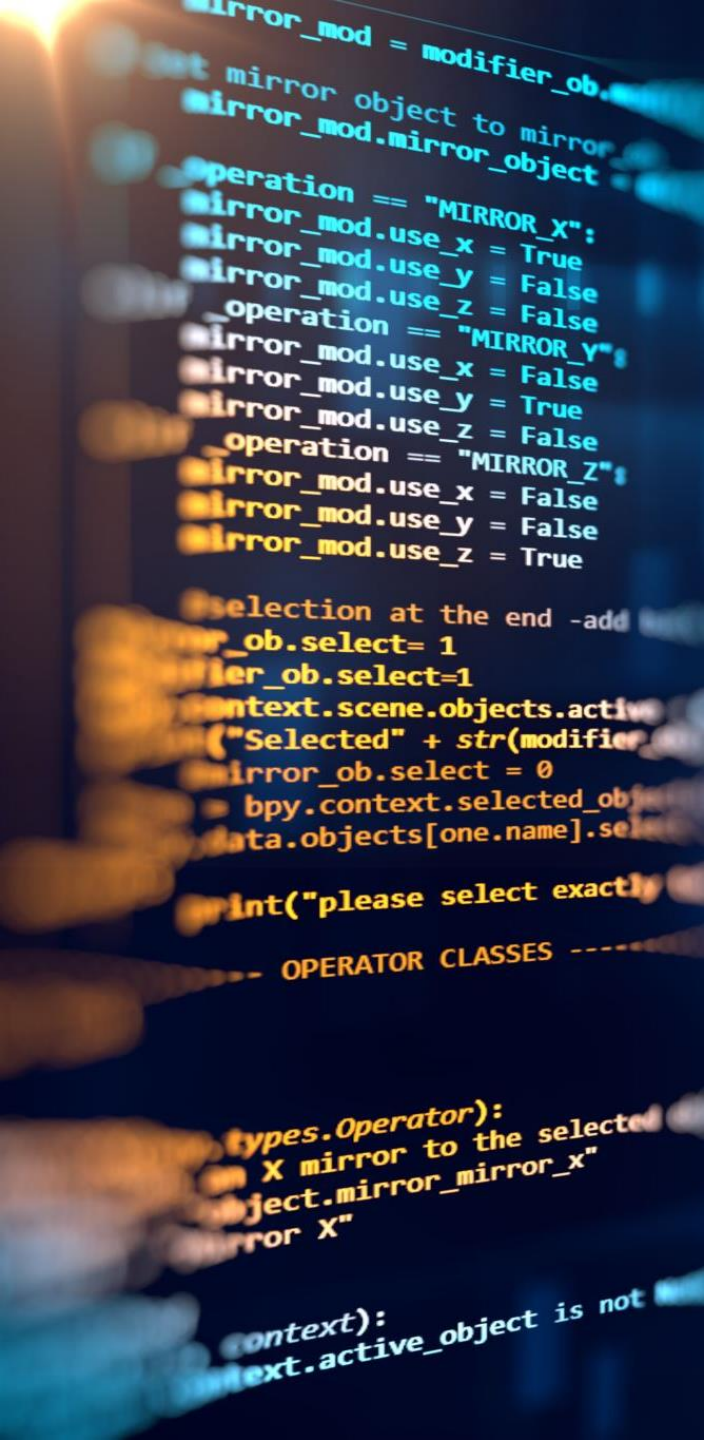
Province: TP. Hồ Chí Minh, Hà Nội, Vũng Tàu

Output:

Street: LTK

Ward: Null

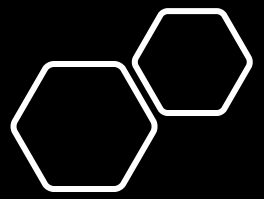
Province: TP. Hồ Chí Minh



1. PROBLEM DEFINITION: Input vs Output

Due to the huge number of Street, for each Province, we choose random 10 Wards for street database

Total: 34 provinces, 34x10 wards, ~34x10x70 streets



1. PROBLEM DEFINI
Input vs Output

Example

Input

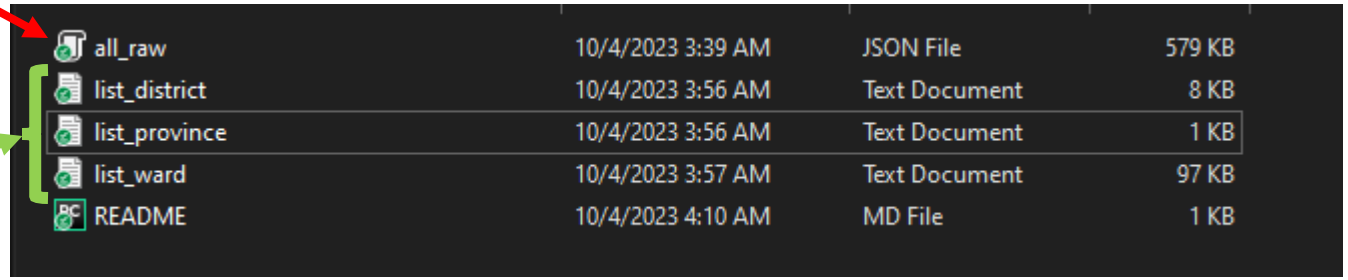
Output

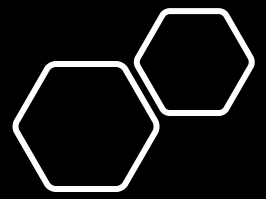

```
[
    {
        "address": "63 Nam Kỳ Khởi Nghĩa, Phường Bến Thành, Thành phố Hồ Chí Minh",
        "result": {
            "street": "Nam Kỳ Khởi Nghĩa",
            "ward": "Phường Bến Thành",
            "province": "Thành phố Hồ Chí Minh"
        }
    },
    {
        "address": "56/33 Nam Kỳ Khởi Nghĩa, Phường Bến Thành, Thành phố Hồ Chí Minh",
        "result": {
            "street": "Nam Kỳ Khởi Nghĩa",
            "ward": "Phường Bến Thành",
            "province": "Thành phố Hồ Chí Minh"
        }
    },
    {
        "address": "69/49/59 Nam Kỳ Khởi Nghĩa, Phường Bến Thành, Thành phố Hồ Chí Minh",
        "result": {
            "street": "Nam Kỳ Khởi Nghĩa",
            "ward": "Phường Bến Thành",
            "province": "Thành phố Hồ Chí Minh"
        }
    },
```

2. DEVELOPMENT & TESTING

Development provided: 1000 samples

Testing provided (private)





2. DEVELOPMENT & TESTING

Score example

name	correct	total	score / 10	max_time_sec	avg_time_sec
Group1	1673	2571	6.51	0.0188	0.006
Group10	1832	2571	7.13	0.0325	0.0068
Group11	1733	2571	6.74	0.0725	0.0045
Group12	1746	2571	6.79	0.5536	0.1399
Group13	1740	2571	6.77	0.0397	0.0041
Group14	1900	2571	7.39	0.0719	0.0132
Group15	1912	2571	7.44	0.0421	0.0093
Group16	1862	2571	7.24	0.0277	0.0042
Group17	1793	2571	6.97	0.1456	0.007
Group18	2271	2571	8.83	0.0076	0.003

3. REQUIREMENTS

General:

- Submit all source code and related data.
- Readme (how to run): The Readme file should be detailed and easily understood.
- The source code will be run on a unique machine without internet.
- **Any copy or cheating will get 0 for the whole course.**

For the exercise:

- Maximum time for 1 request ≤0.1 s
- Average time for 1 request: ≤0.01 s
- Duration: 3 weeks (a part of private data will be released 2 days before the deadline
- Support 01 free test.
- Do not apply a machine learning approach, just algorithms.
- **Score= 80% private test + 20% public test**


Q&A